

Tejas Satish Navalkhe

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Professional Summary

AI/ML and Software Engineer with over **3 years of experience** building scalable machine learning solutions for fintech, transport, and edtech. I specialise in **Machine Learning, Deep Learning, NLP, & MLOps**, with a track record of driving impact: delivered a **19.32% revenue boost** via dynamic pricing and cut trading system latency by **40%**. Hold a **Distinction in MSc Data Science (AI Specialisation)** from Newcastle University, UK, and excel in deploying high-impact projects with **Python, TensorFlow, PyTorch, and cloud platforms**.

Work Experience

Deputy Manager (Technology)

August 2025 – Present

Jalgaon Janata Sahakari Bank Ltd.

Jalgaon, Maharashtra, India

- **Automated MIS and RBI regulatory reporting** across **10+ departments** using **Python, SQL, FastAPI, Pandas, Matplotlib, and Seaborn**, reducing manual effort by **70%**, reporting errors by **90%**, and ensuring timely regulatory submissions.
- **Automated AML/KYC compliance workflows** by developing a one-click tracking and reporting solution for **Internal Accounts Monitoring, Enhanced Due Diligence (EDD), and Suspicious Transaction Reports (STR)**, generating **11 FIU-IND reporting formats** and reducing report preparation time from **3 days to under 10 minutes**, significantly improving reporting accuracy and regulatory compliance.
- **Designed and deployed secure on-premises automation infrastructure** for report generation, data storage, and retrieval, ensuring compliance with **RBI cybersecurity and data governance guidelines**.
- **Built internal banking automation platforms**, including a **Complaint Tracking System, Daily Transaction Monitoring System, and Technical Decline Reporting System** for ATM, UPI, ONUS, and digital banking channels, improving operational communication efficiency by **80%**.

Freelance AI/ML Engineer

September 2024 – June 2025

Self Employed

Jalgaon, Maharashtra, India

- Created **AI/ML solutions for 15+ edtech/fintech clients**, including a real-time trading system that slashed latency by **30%** using Python, Flask, Redis, and WebSocket. Built scalable REST/WebSocket APIs (FastAPI) on AWS, cutting deployment time by **25%** while handling **5,000+ concurrent users**.
- Enhanced **NLP and course recommendation engines** (BERT, GPT) for an edtech platform, lifting user engagement by **25%**. Systematised retraining and CI/CD (GitHub Actions) saved **15 hours/month** and ensured **99.8% uptime**.
- Boosted model accuracy by **12%** via orchestrated hyperparameter tuning (MLflow). Streamlined data pipelines (Apache Airflow) and web scraping accelerated data aggregation by **50%** for real-time predictions.

Data Science Intern

May 2024 – September 2024

Co Wheels Car Club

Newcastle Upon Tyne, United Kingdom

- Designed a **dynamic pricing model** using ML, analysing data from **30,000+ users** and **600+ vehicles** (including 200+ EVs). Optimised pricing for location, peak hours, and seasonality, driving a **19.32% revenue increase** versus static models.
- Created a **Flask API simulation tool** for real-time pricing adjustments. Automated preprocessing of **40+ features** cut manual work by **40%** and improved model accuracy.
- Synced data with Co Wheels' booking system, reducing data sync issues by **25%**. Insights guided their shift to new pricing tool, boosting profits without compromising user satisfaction.

Freelance Software Engineer

September 2022 – April 2024

Finvestor Capital

Remote, India

- Architected an **algorithmic trading system** enabling **sub-millisecond execution**, increasing trading volume by **20%** and returns by **10%** in 3 months. Deployed on AWS/Docker for **50% faster scaling**, with Redis caching speeding data retrieval by **35%** and **99.9% uptime**.
- Formulated a **trading platform** with risk management. Selenium-based broker logins cut trade execution by **30%**; Plotly dashboards improved real-time analytics accuracy by **25%**.
- Used **Redis pub/sub** to streamline multi-server data flow and refined SQL databases, cutting prediction latency by **20%** during high-frequency trading.

Education

MS in Data Science (With Specialisation in Artificial Intelligence) **September 2023 – November 2024**

Newcastle University, Newcastle Upon Tyne, United Kingdom

Grade: Distinction

- **Modules:** Machine Learning, Deep Learning, Big Data Technology, Statistics, EDA, Data Management.

B. Tech in Computer Science Engineering

August 2019 – June 2023

Prestige Institute Of Engineering, Management & Research, Indore, India

CGPA: (8.78/10)

- **Modules:** Data Structure, Cloud Computing, Project Management, Machine Learning, DBMS, Software Engineering.

Projects

Scalable AI-Driven Algorithmic Trading System with Real-Time Automatic Trading

- Constructed a low-latency trading system using Apache Spark Streaming and WebSockets, reducing data delay by 40% and integrating AI models (LSTM, ARIMA) to improve trade accuracy by 15% via real-time sentiment analysis.
- Containerised deployment with **Docker on AWS**, achieving **99.9% uptime**, and designed a custom risk-management module to limit losses to **2% per trade** while optimizing returns.
- Reverse-engineered **AngelOne/Zerodha APIs** to unify order execution across brokers and implemented a Selenium-based testing framework, cutting execution bugs by **30%** and ensuring compliance with trading regulations.

AI-Driven Face Recognition Attendance System

- Developed a **face recognition-based attendance system** using **OpenCV**, automating attendance tracking and reducing manual effort by **70%**.
- Achieved an impressive **99.38% recognition accuracy**, ensuring **reliable and consistent identification** under varied real-world conditions.
- Integrated **Amazon S3** for **large-scale image storage**, enabling **scalable, cloud-ready deployment** with **fast and efficient data access** for real-time face matching.

Skills

Programming Languages: Python, SQL, JavaScript

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, CatBoost

Natural Language Processing: Transformers, Hugging Face, spaCy, OpenAI APIs, BERT, GPT

MLOps: MLflow, Weights & Biases (W&B), DVC, Airflow, Docker, Kubernetes

Cloud & DevOps: AWS, Docker, CI/CD (GitHub Action), Monitoring & Logging

Software Development & Testing: Clean Code Principles, Design Patterns, Unit Testing (Pytest), Integration Testing

Version Control & Collaboration: Git, GitHub

Data & Databases: MySQL, PostgreSQL, MongoDB, Redis

Web Technologies & API: RESTful API, WebSocket, FastAPI, Flask

Data Engineering & Processing: Pandas, NumPy, Apache Spark, ETL Pipelines

Visualisation: Matplotlib, Seaborn, Plotly, Power BI

Awards & Achievements

NCL+ Award

Newcastle University

May 2024

Newcastle Upon Tyne, United Kingdom

Best Student Of the Batch (2019-2023) Award

Prestige Institute Of Engineering, Management & Research

April 2023

Indore, India

Key Certifications

- **International Conference on Cognition, Human and Artificial Intelligence** by KBCNMU — January 2026
- **Smart Banking with AI – A Deep Dive for Bankers** by Indian Institute of Banking & Finance — September 2025
- **Machine Learning with Scikit-Learn** by Madecraft — August 2024
- **Data Science & Advance Machine Learning using Python** by Grras Solutions — January 2022

Publications

- **“Design and Development of Smart Virtual Assistant Using Latest Tools and Technologies”,**
International Journal of Core Engineering & Management, Feb. 2022, ISSN: 2348-9510